High-Throughput Sequencing

Introduction to Bioinformatics · NGS day 1

Amrei Binzer-Panchal

Welcome

High-throughput sequencing: from the molecule to the reads.

With slides based on presentations of Nicolas Delhomme and Prof. Nathaniel Street

Deciphering the blueprint of an organism

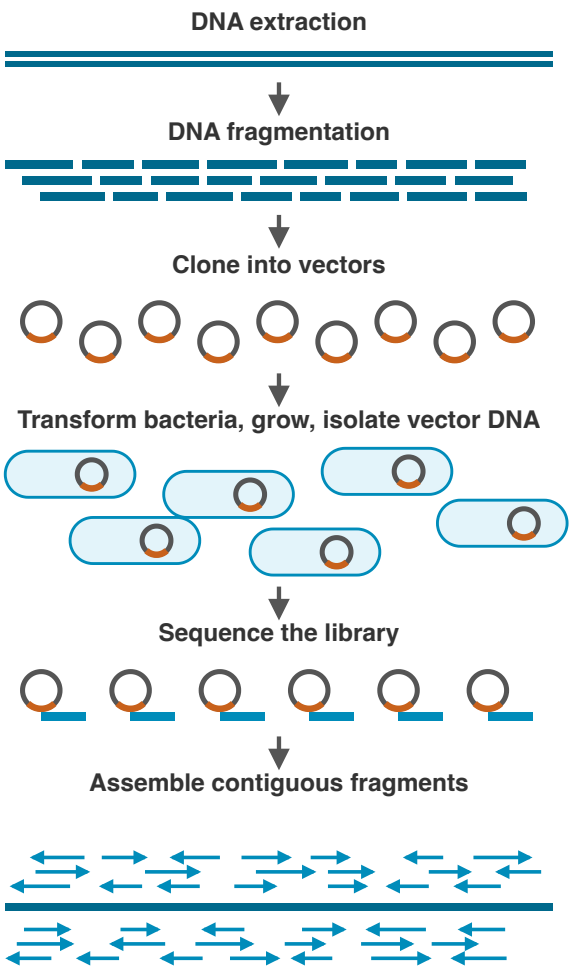
- **1977:** Sanger's chain-termination method, one fragment per reaction
- **2001:** first draft human genome (Human Genome Project and Celera). Full project ran 1990–2003: ~13 years, ~$2.7 billion
- **2022:** first complete human genome (T2T-CHM13, gapless except chrY), made with long reads
- **Today:** a few hundred dollars in reagents and a 1–2 day run

Lander et al. 2001 Nature 409:860; Venter et al. 2001 Science 291:1304; Nurk et al. 2022 Science 376:44; genome.gov HGP cost fact sheet; Shendure et al. 2017 Nature 550:345

This lecture

1. **Sanger sequencing:** the beginning
2. **Library preparation:** DNA and RNA
3. **Short reads:** Illumina and competitors
4. **Long reads:** PacBio and Oxford Nanopore
5. **Choosing a platform:** errors, what's new, NGI

Sanger sequencing: how it started

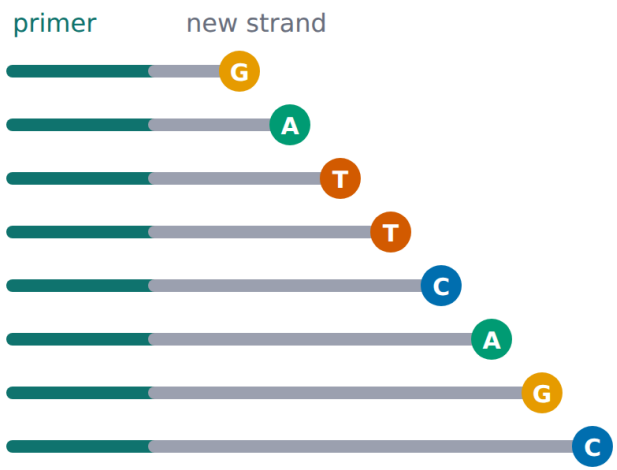


Own work, CC BY 4.0

- One reaction per DNA fragment
- Genomes needed clone libraries: lots of lab work, lots of chances for mistakes
- Low throughput: one read (700–1000 bases) per capillary, 96 capillaries per run
- Still used today to check single loci, clones and PCR products

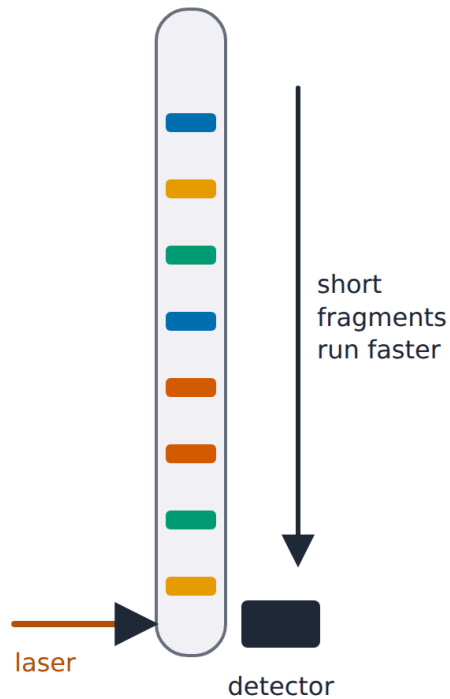
Chain termination

1 Chain-termination reaction

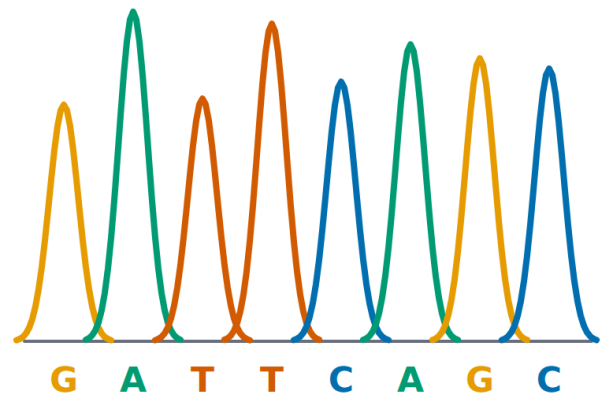


Normal dNTPs plus a little dye-labelled ddNTP. A ddNTP has no 3′-OH, so each copy stops at a random position and its last base carries the colour.

2 Size separation



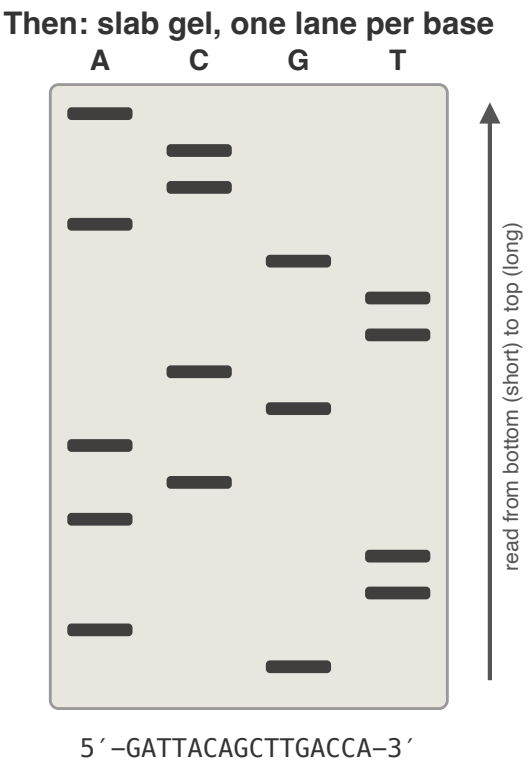
3 Read-out: chromatogram

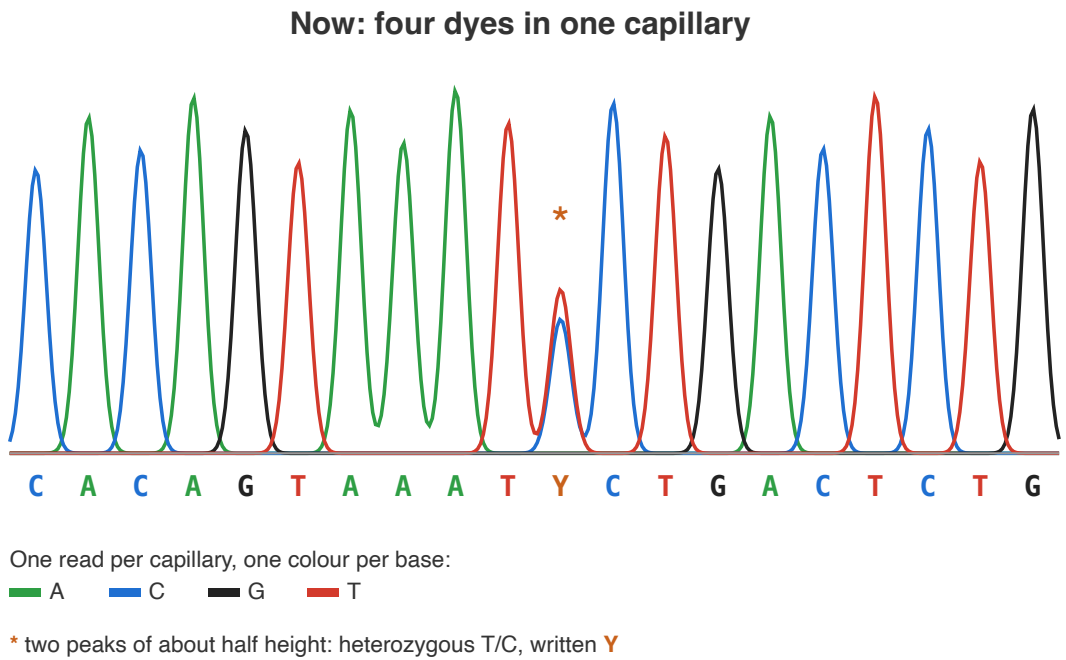


One read per capillary, up to ~1,000 bp, very accurate. Still the standard for checking single loci and clones.

Own work, CC BY 4.0

Reading a Sanger trace



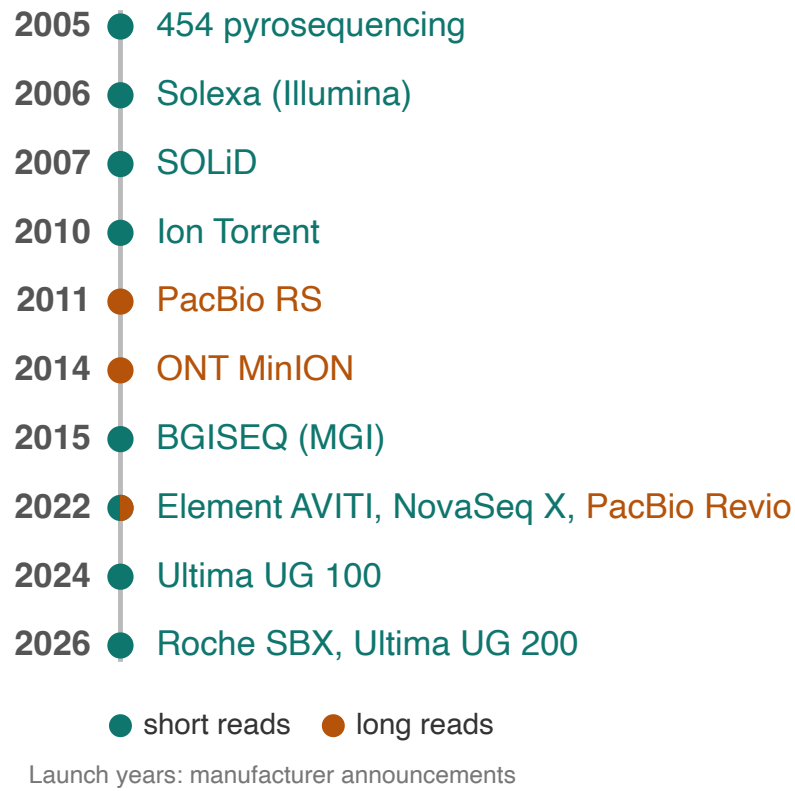


Own work, CC BY 4.0

- **Then:** radioactive slab gels, one lane per base, read by eye
- **Now:** four dyes in one capillary, read by a laser
- Two overlapping peaks of about half height = a heterozygous position

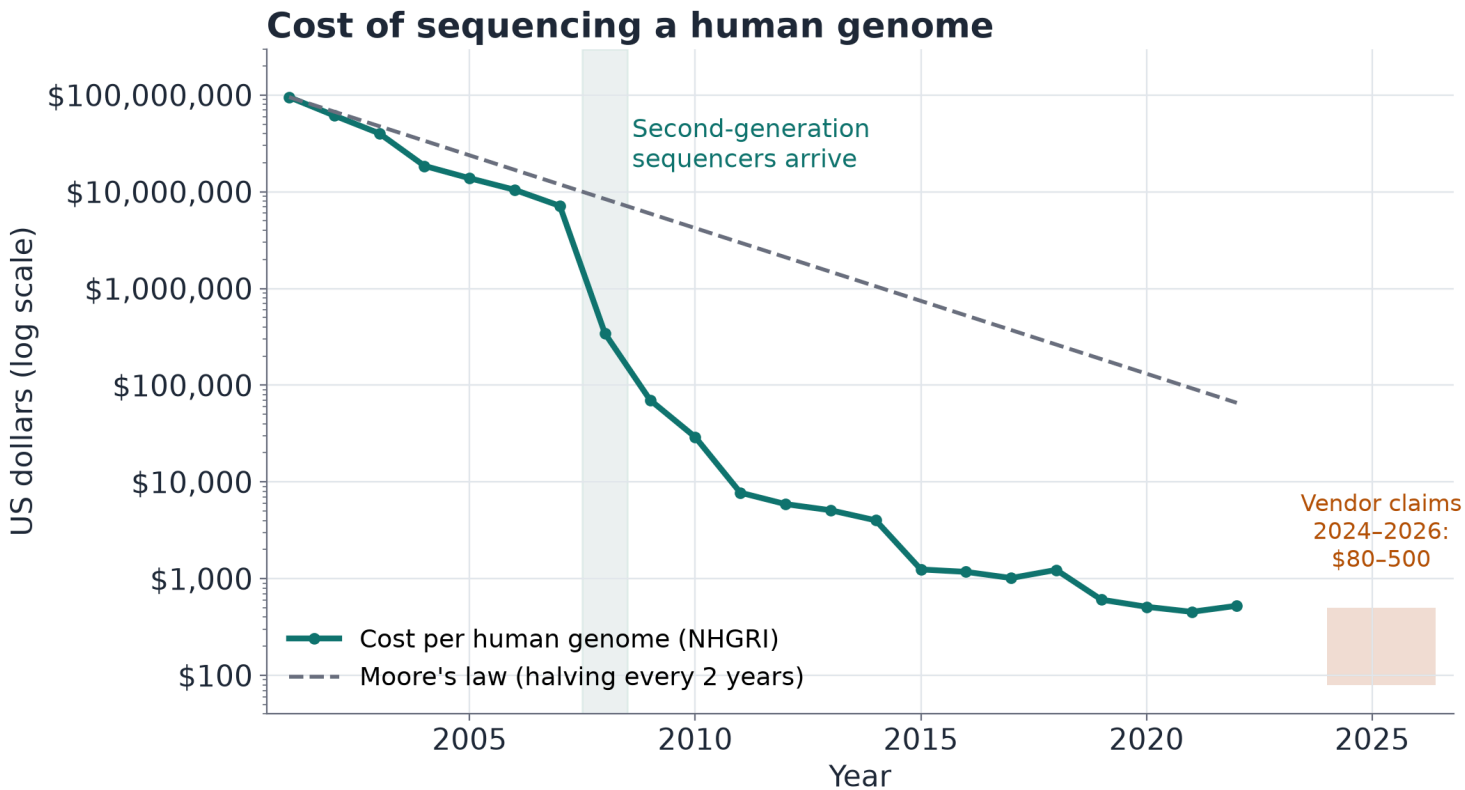
"Next" (2nd) generation sequencing

- Sequencing-by-synthesis: massively parallel sequencing
- Millions to billions of fragments sequenced at once
- Illumina still dominates, but there is real competition again: Element AVITI, Ultima UG 100/200, MGI / Complete Genomics DNBSEQ, Roche SBX
- Gone: Roche 454, SOLiD, Helicos. Ion Torrent: clinical niche



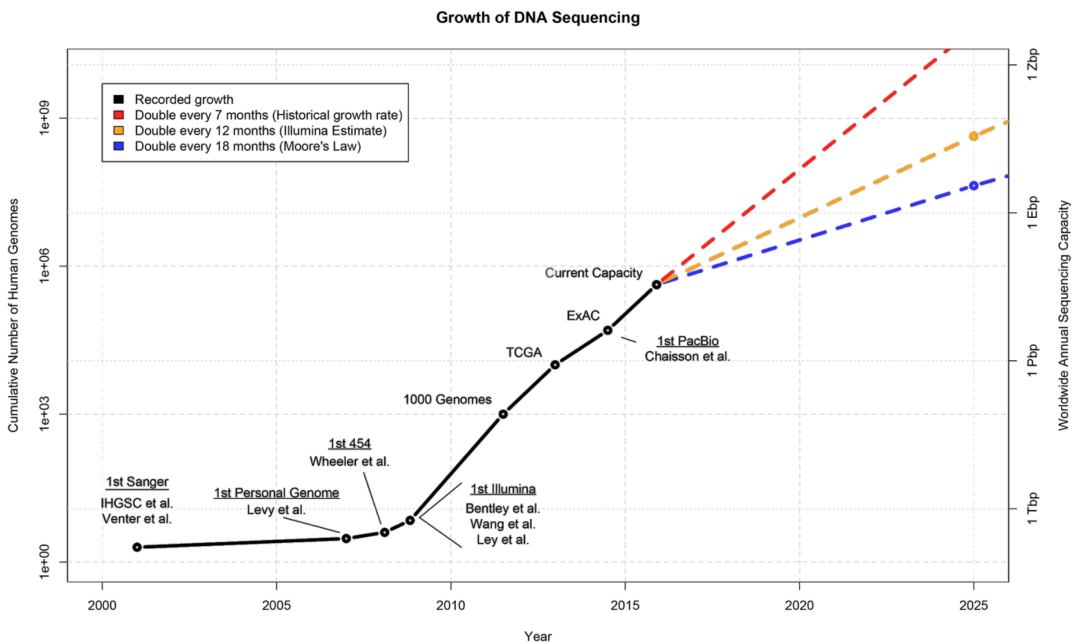
Launch years: manufacturer announcements

The $1000 genome



Data: NHGRI via Our World in Data (CC BY 4.0). 2024–2026: manufacturer claims. Figure: own work, CC BY 4.0

Sequencing capacity grows faster than Moore's law

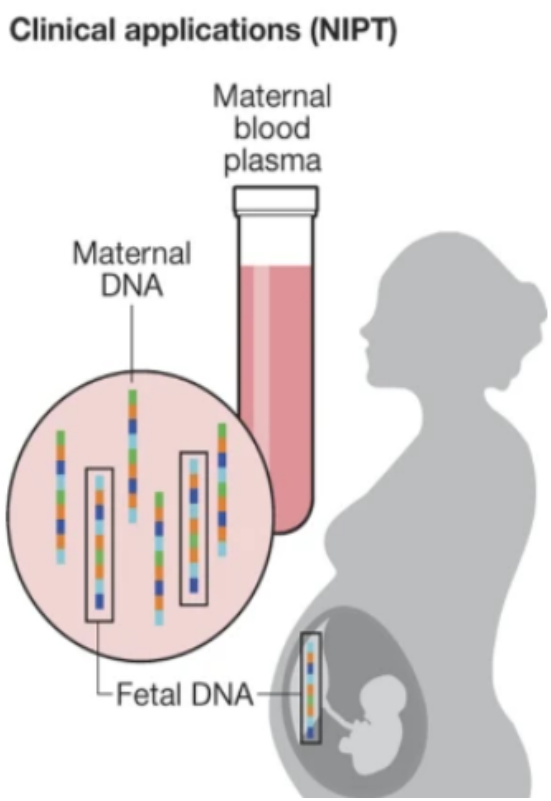


- Computing power per dollar has kept roughly doubling every two years
- Sequencing got cheaper, but computers do not keep up: storing and analysing the data becomes as hard as producing it
- 2015 projection: 100 million to 2 billion genomes by 2025. Reality: a few million

First human genome: 13 years.
SciLifeLab 2016: 1 h

Stephens ZD et al. (2015) Big Data: Astronomical or Genomical? PLoS Biol 13:e1002195 (CC BY 4.0)

What can we do with it?



- **Non-invasive prenatal testing:** fetal DNA in the mother's blood is sequenced
- The film *Gattaca* (1997) imagined genetic profiling at birth, less than 30 years ago

NIPT panel: Shendure et al. (2017) DNA sequencing at 40, Nature 550:345 (© Springer Nature)

Preparing DNA and RNA

Turn the sample into a library

From sample to library



Own work, CC BY 4.0

- The extraction method decides what is possible later: purity, fragment length, RNA integrity
- Long reads need high-molecular-weight DNA: no vortexing, no PCR; HiFi sheared to 15–20 kb, ultra-long not sheared

Fragmenting DNA

Mechanical



Sonication or focused acoustic shearing

Random breaks, needs an instrument

Enzymatic



Endonuclease mix cuts double-stranded DNA

Fast and cheap, small sequence bias

Tagmentation



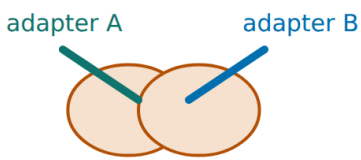
Tn5 cuts and adds adapters in one step

Fewest steps, slight insertion-site bias

Own work, CC BY 4.0

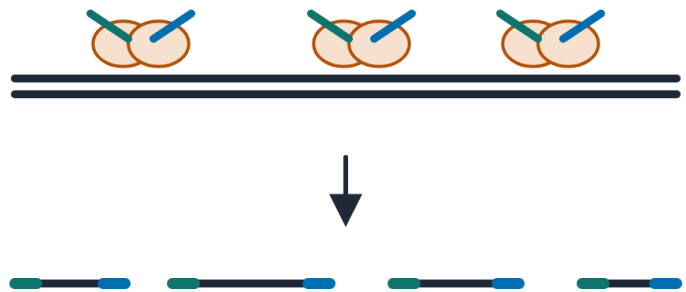
Tagmentation

1 Transposome



Tn5 transposase (two subunits) loaded with two adapter sequences

2 Cut and insert in one step



DNA is fragmented and tagged with adapters at the same time

3 Short PCR



Primers on the adapters add sample indexes and the flow-cell sequences

Replaces shearing, end repair, A-tailing and adapter ligation. With bead-linked transposomes (e.g. Illumina DNA Prep, formerly Nextera DNA Flex) the fragment size depends less on DNA input.

Own work, CC BY 4.0

From RNA to a library

1 Pick the RNA

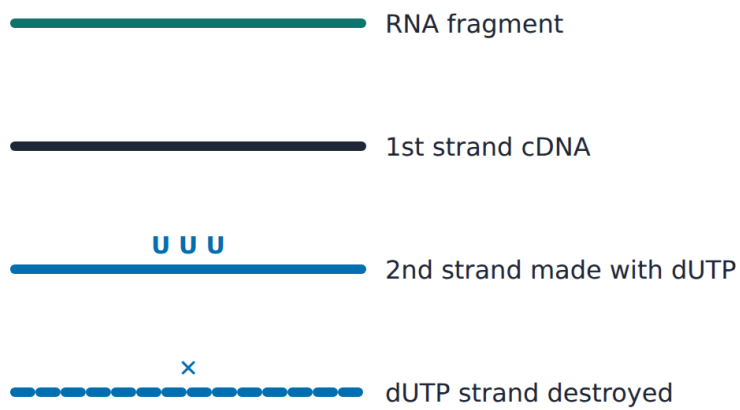
total RNA: ~80–90% is rRNA



Poly-A selection
oligo-dT beads catch the A tail.
mRNA only; needs intact RNA.

rRNA depletion
Probes remove rRNA. Keeps non-poly-A RNA; works on degraded RNA.

2 Copy to cDNA, keep the strand



Only the first strand is sequenced, so each read tells which DNA strand the gene was transcribed from: a stranded library.

3 Library



cDNA insert

Adapters, indexes and PCR as for DNA. Then sequence: usually short reads, paired-end or single-end.

Own work, CC BY 4.0

Read types

Single-end

one read from one end of each fragment



Paired-end

both ends of the same fragment; insert size roughly known



Long read

the whole molecule in one read: 10 kb to over 1 Mb



UMI

a random tag added before PCR marks the original molecule



3 reads, same UMI

= 1 original molecule

Historical: mate-pair libraries and 10x linked reads (discontinued 2020), replaced by long reads and Hi-C.

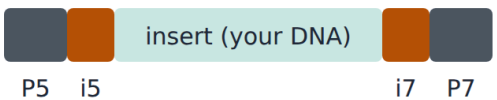
Own work, CC BY 4.0

Short-read sequencing

Illumina sequencing by synthesis, and instruments

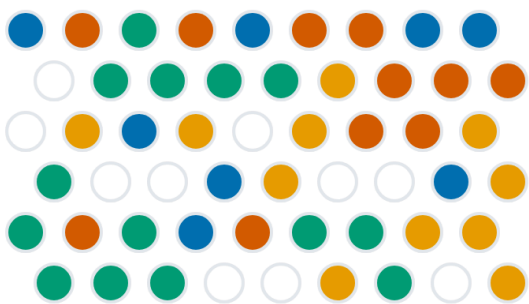
Illumina: sequencing by synthesis

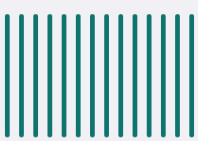
1 Library molecule



P5/P7 bind the flow cell;
i5/i7 indexes identify the sample

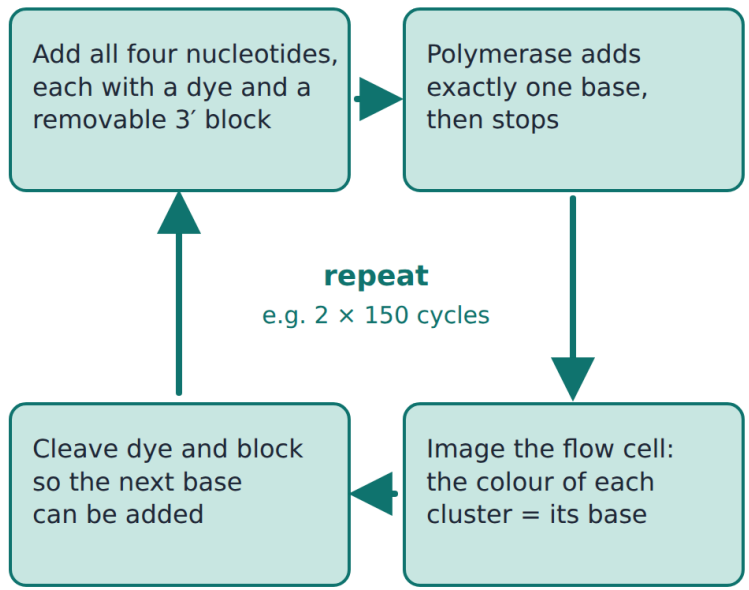
2 Clusters on a flow cell

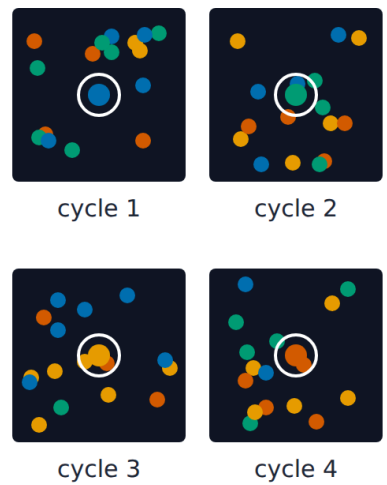




Each nanowell holds ~1,000 identical copies of one molecule, made by on-surface amplification

3 Sequencing by synthesis: one base per cycle





The circled cluster reads: **C A G T …**

Own work, CC BY 4.0

Illumina sequencing by synthesis

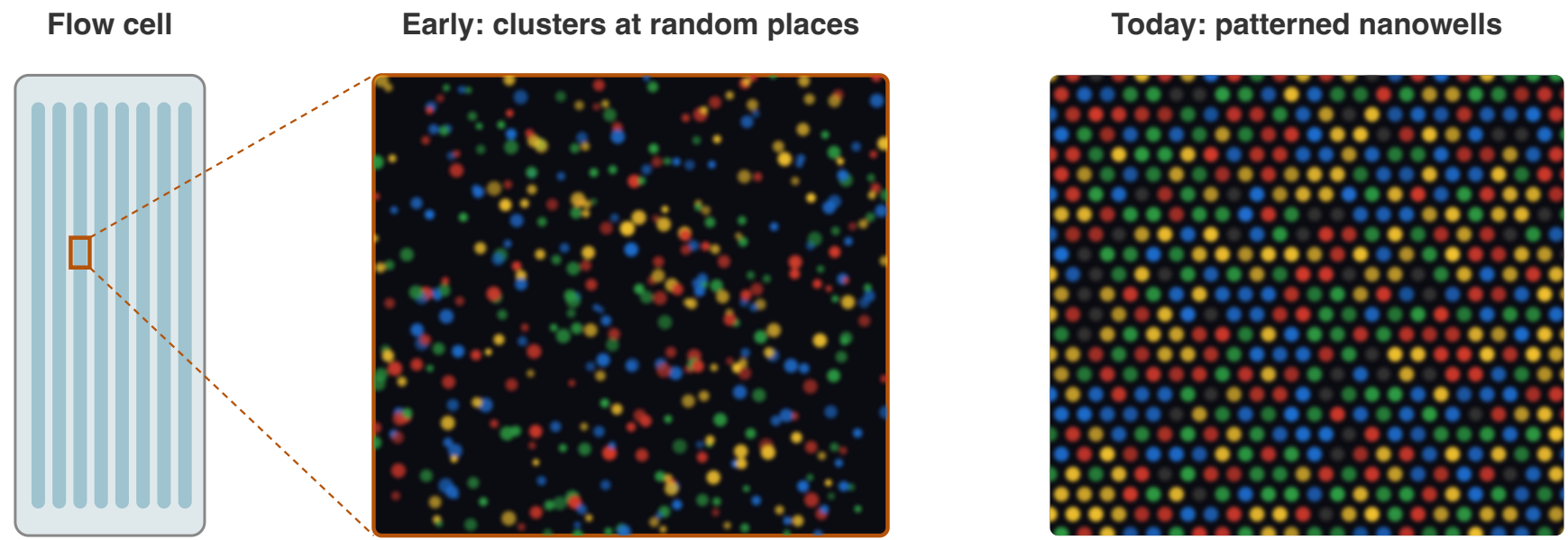
Watch the following video to understand Illumina's sequencing by synthesis:

▶ **Overview of Illumina Sequencing by Synthesis Workflow**

Illumina, on YouTube

Video: © Illumina, Inc.

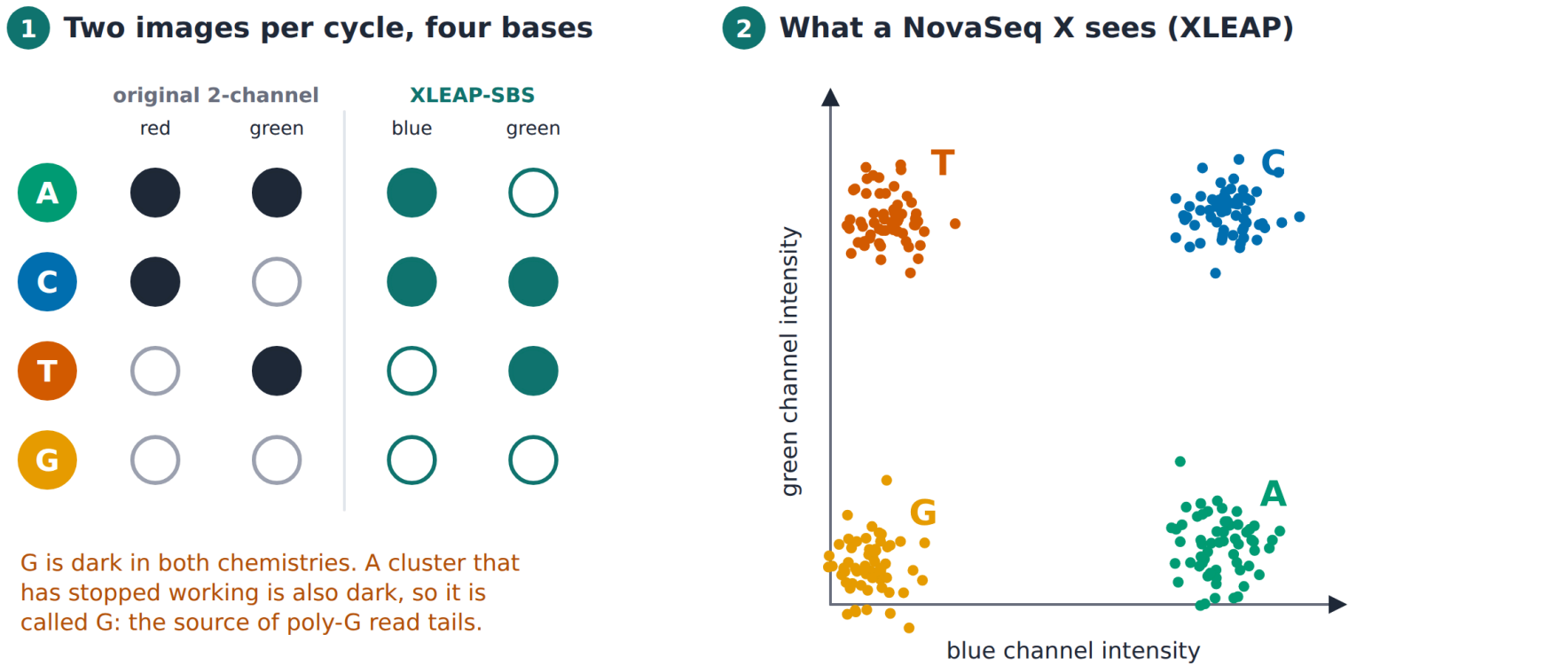
This is how it looks



Own work, CC BY 4.0

- Lanes are imaged tile by tile
- Each spot is one cluster; colour = base in this cycle
- Early flow cells: clusters at random positions. Today: patterned nanowells, billions of clusters

Improvements: two-colour chemistry



Original 2-channel: NextSeq 500/550, NovaSeq 6000. XLEAP-SBS: NovaSeq X, NextSeq 1000/2000, MiSeq i100.
4-channel (HiSeq, classic MiSeq): one dye per base. 1-channel (iSeq 100): one dye, two chemistry steps.

Own work, CC BY 4.0. Encoding: Illumina Knowledge 000008422

Figure: own work, CC BY 4.0; colour encoding: Illumina Knowledge article 000008422

Long-read sequencing

Third generation: single molecules, long reads

	PacBio HiFi	Oxford Nanopore
How	Polymerase copies a circular template, watched in real time	Strand pulled through a protein pore, current is measured
Length	15–20 kb	As long as the molecule, up to > 1 Mb
Accuracy	Typically Q30 (99.9%)	~Q26 (99.75%) per read

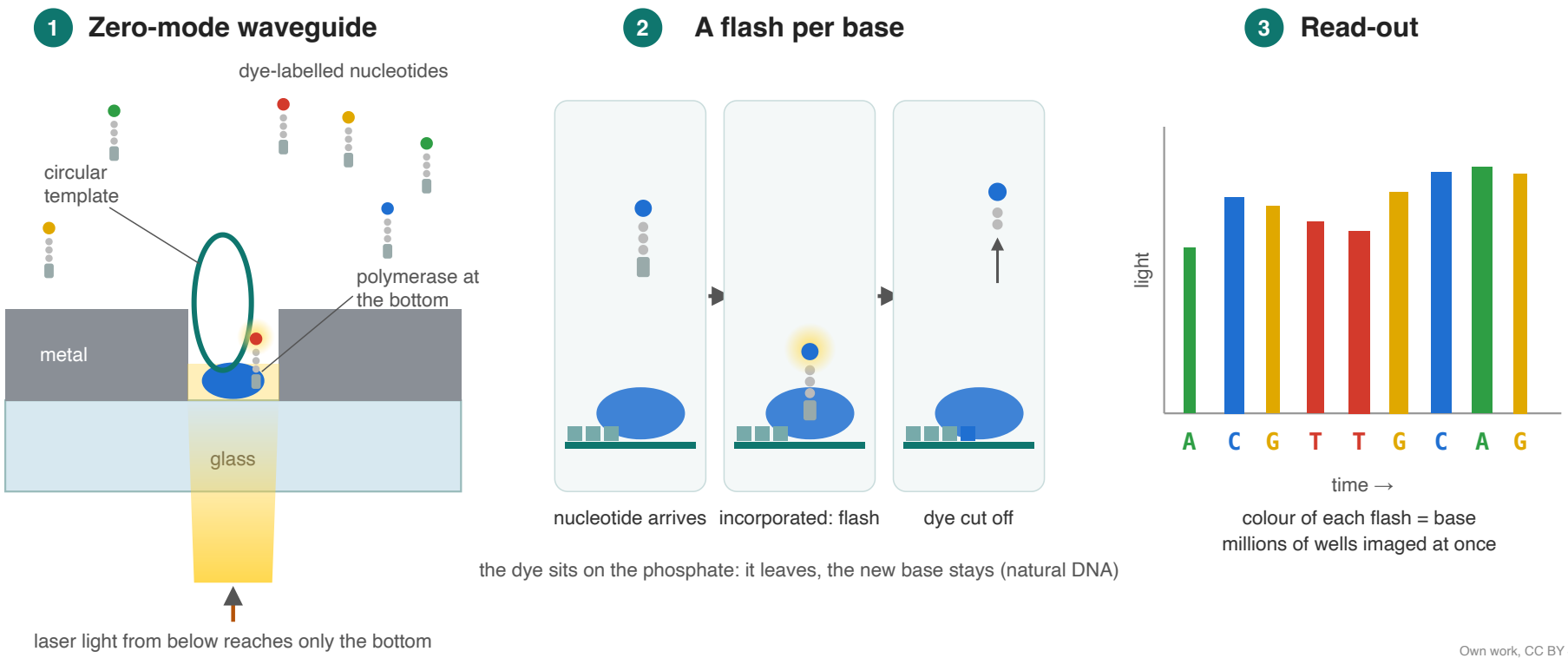
Both: no PCR needed, and methylation is read directly from the native molecule.

Note

- Q20 = 1 error in 100 bases
- Q30 = 1 in 1,000
- Q40 = 1 in 10,000.

Accuracy: manufacturer specifications (PacBio 2024, Oxford Nanopore 2025)

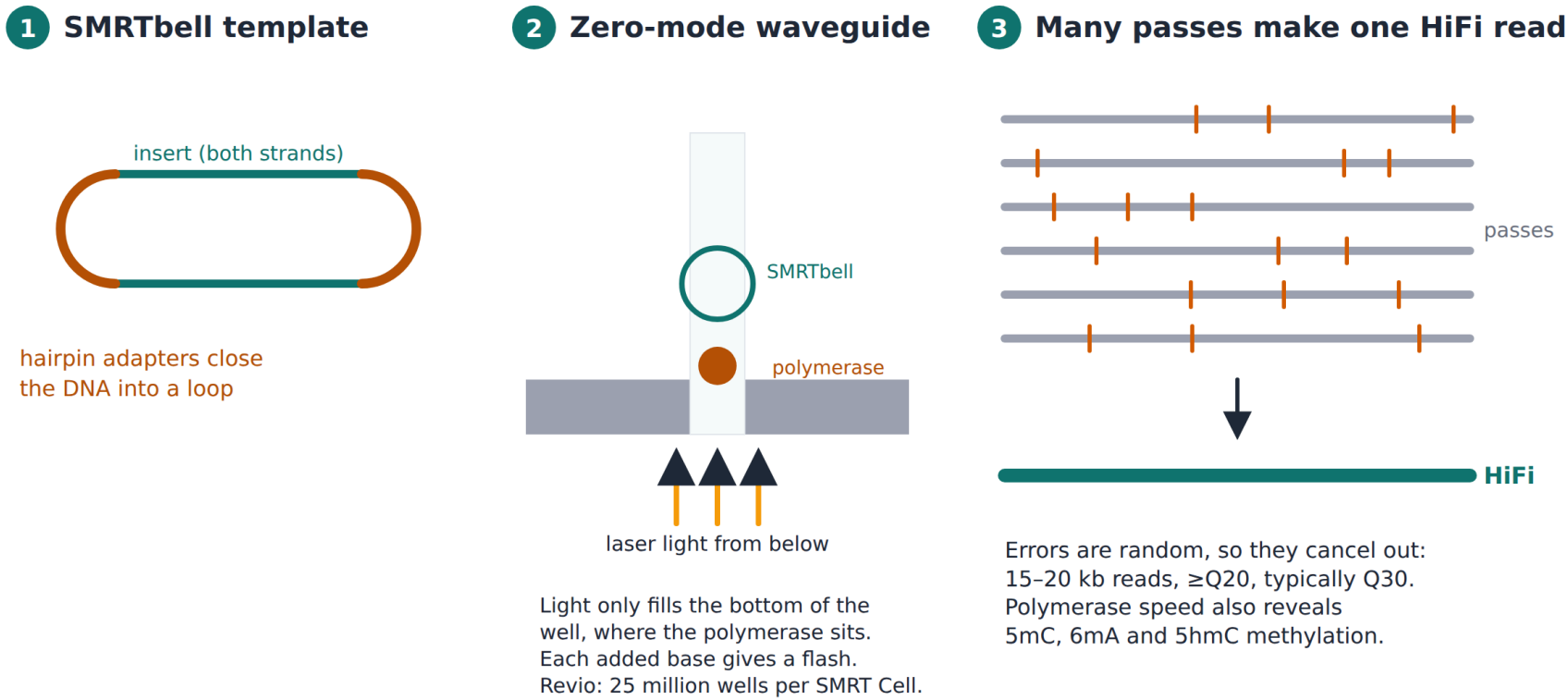
PacBio: single-molecule real-time sequencing



Own work, CC BY 4.0

- A polymerase sits at the bottom of a zero-mode waveguide
- Dye on the phosphate: a flash per added base, then the dye is cut off
- Revio: 25 million wells per SMRT Cell

PacBio HiFi: accuracy from repetition



Own work, CC BY 4.0

Figure: own work, CC BY 4.0. Revio SPRQ-Nx: ~$345 list, < $300 at scale per HiFi human genome (PacBio press release, 26 May 2026)

Oxford Nanopore

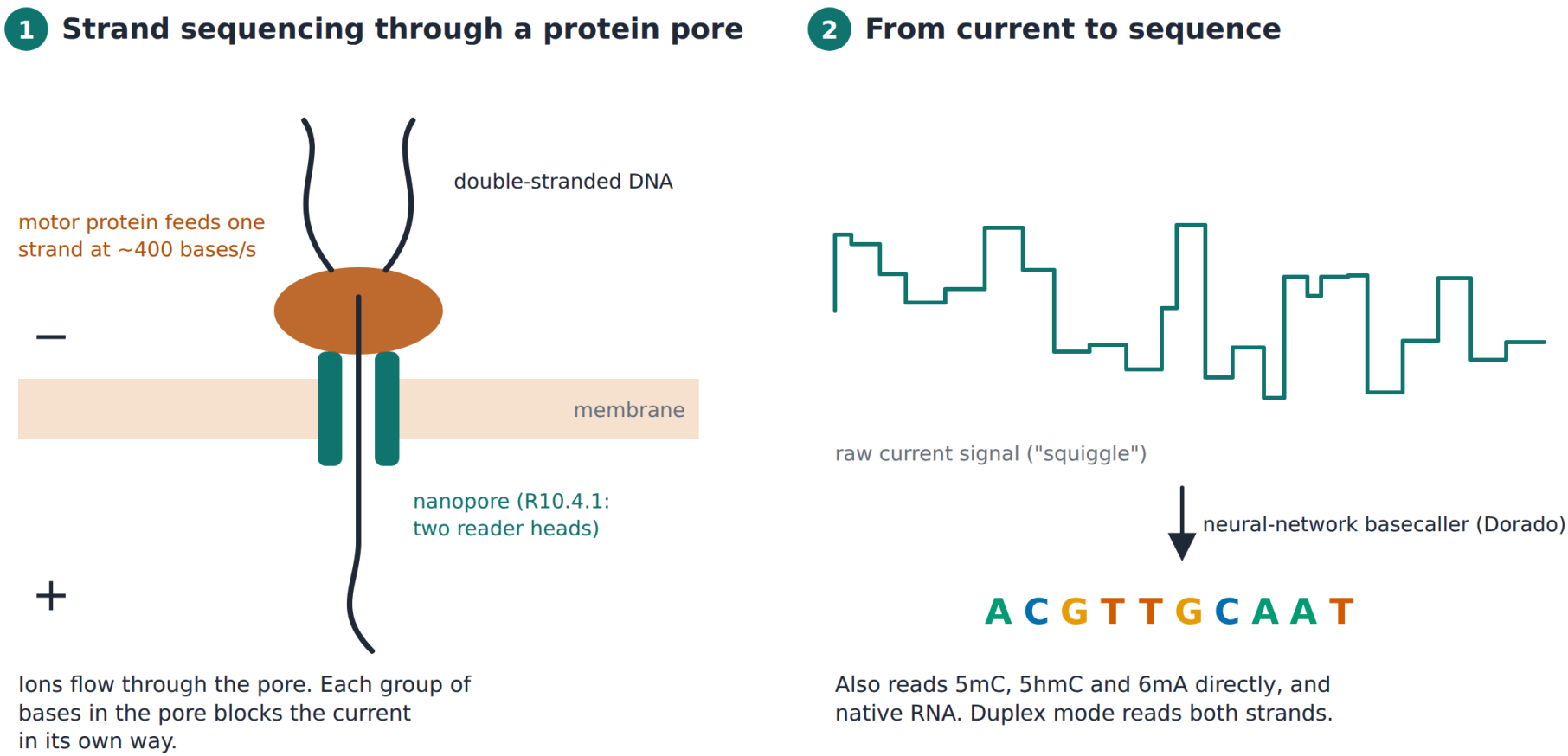


Figure: own work, CC BY 4.0. Accuracy: nanoporetech.com/platform/accuracy (accessed Oct 2026)

Long reads for RNA: whole transcripts



Short reads

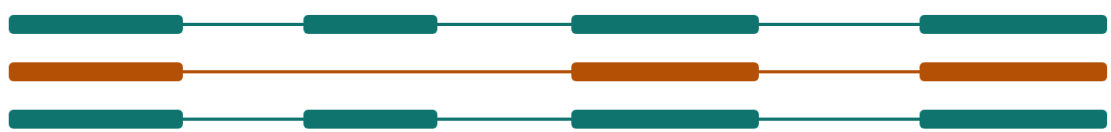
Reads inside shared exons fit both isoforms.



which isoform?
only reads in isoform-specific parts tell: exon 2 (teal) or the skip junction (orange)

Long reads

One read = one transcript.

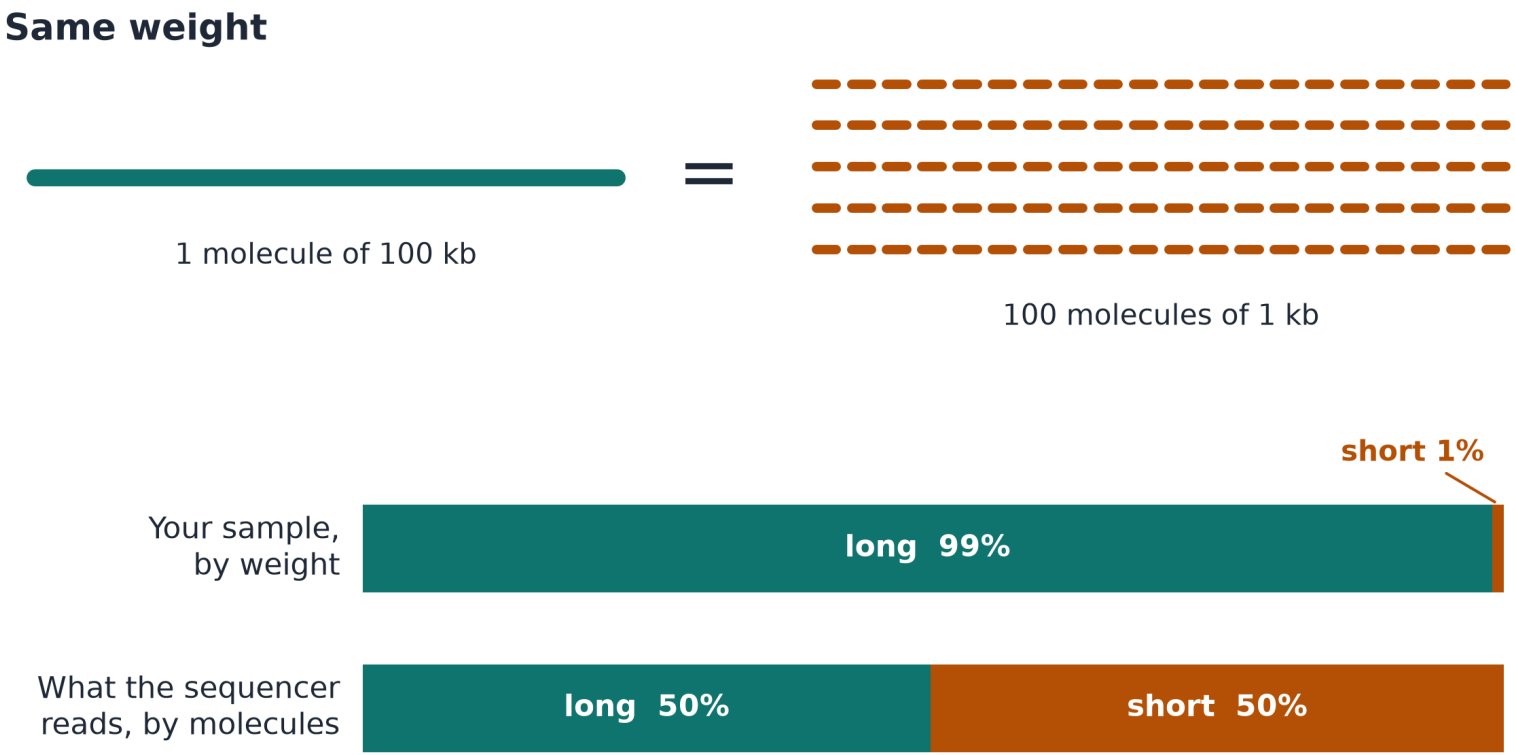


A, B, A

PacBio Kinnex / Iso-Seq (full-length cDNA) · ONT cDNA or direct RNA (also reads RNA modifications)

Own work, CC BY 4.0

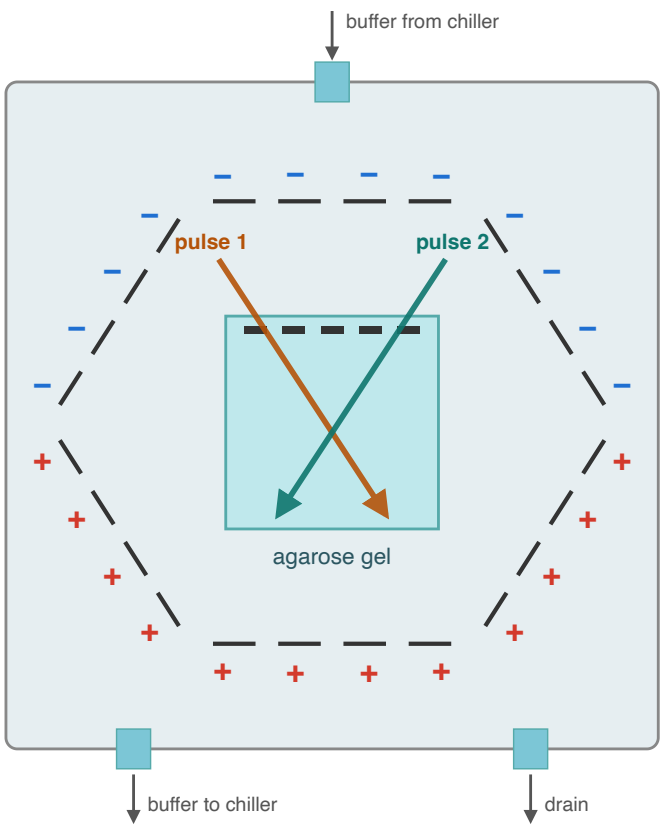
Short fragments can take over a long-read run



Example: 99% of the DNA mass is 100 kb molecules, 1% is 1 kb fragments. Figure: own work, CC BY 4.0. Idea after the Albertsen lab blog.

Figure: own work, CC BY 4.0. Idea after a post on the Albertsen lab blog (Aalborg University; no longer online)

Getting long DNA: pulsed-field gel electrophoresis

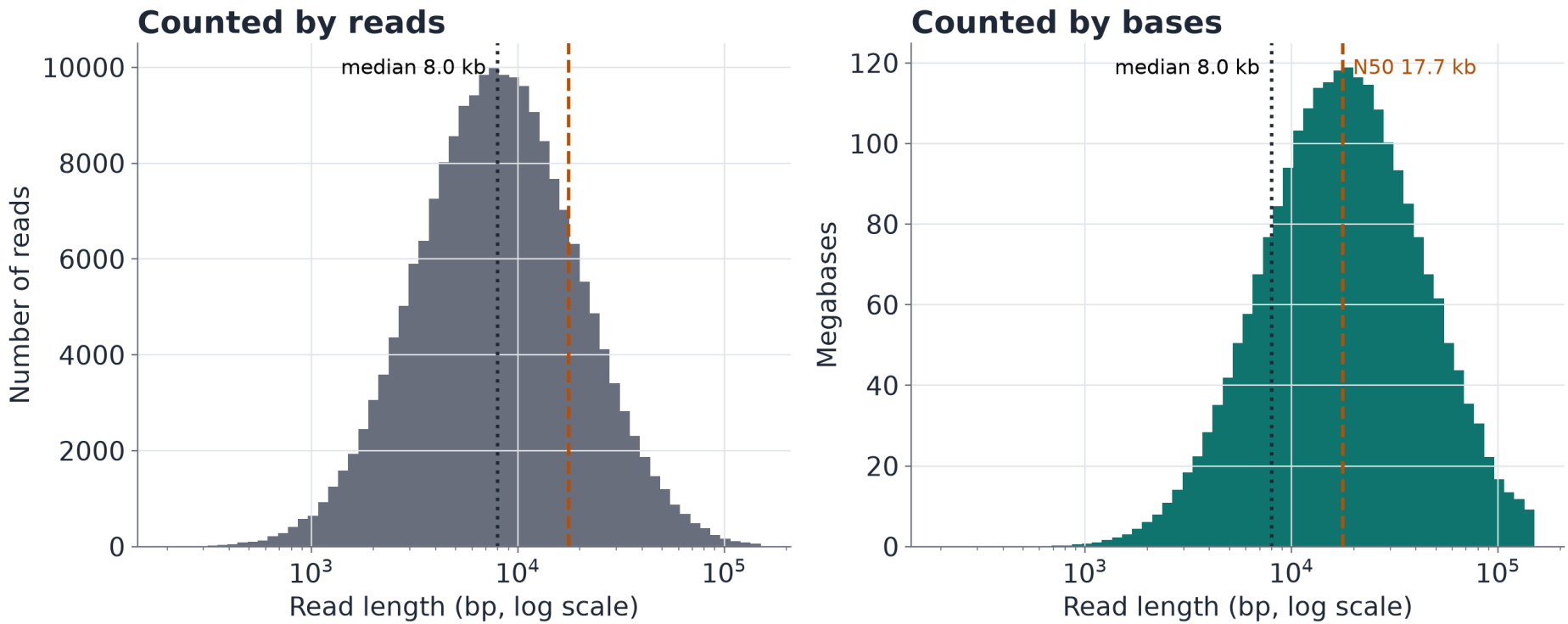


Own work, CC BY 4.0

The field switches direction every few seconds.
Long DNA turns more slowly, so it separates by size.

- Gentle extraction: high-molecular-weight DNA
- Fragments > 15–20 kb run together on a normal gel: use pulsed-field gels
- Remove short fragments: gel-based (BluePippin, cut-off ~10–15 kb) or bead-based kits

Read-length distributions and N50



Simulated long-read run (log-normal lengths). Most reads are short, but most bases sit in long reads. Own work, CC BY 4.0.

N50: half of all bases are in reads at least this long.

Simulated data. Figure: own work, CC BY 4.0

Choosing a platform

Strengths and typical errors

Which platform for which question?

	Illumina NovaSeq X	Element AVITI	PacBio Revio (HiFi)	Oxford Nanopore
Read length	2 × 150 bp	2 × 150 bp	15–20 kb	As long as the molecule: ~100 bp to > 1 Mb
Accuracy	Q30+	Q30+ (Q40+ with UltraQ)	Q20+, typically Q30	Q20–Q26 per read (duplex Q30)
Best for	Large cohorts, RNA-seq counts, lowest cost per base	Very accurate short reads	Genome assembly, phasing, full-length isoforms	Longest reads, structural variants, direct RNA, field work

Short reads are cheap and plentiful, long reads span repeats and whole transcripts.

Manufacturer specifications, 2024–2026

Typical errors and where they come from

Homopolymers. AAAAAAA miscounted → false insertions or deletions. Worst on ONT.
Not in FastQC: shows up in variant calls

Duplicates. PCR copies inflate coverage. Normal for highly expressed genes.
FastQC: sequence duplication levels

GC bias. GC-rich or GC-poor regions get fewer reads.
FastQC: per sequence GC content

Index hopping. Reads land in the wrong sample.
Prevent with unique dual indexes

Poly-G and adapters. "No signal" is called G; short inserts read into the adapter.
FastQC: adapter content → trim

Artefact or biology? Think of the technical explanation first.
→ *QC lecture and today's exercises*

Höjer et al. (2026) NAR Genom Bioinform 8:lqag053; Illumina and Oxford Nanopore technical notes

News and outlook

New developments and sequencing in Sweden

What's new in 2026

- **Roche AXELIOS 1 / SBX (June 2026):** DNA copied into a ~50× longer polymer, read through a nanopore; same-day genomes claimed (research use)
- **Ultima UG 200 (Feb 2026):** open spinning wafer instead of a flow cell; twice the output of the UG 100, half the run time
- **Element VITARI (H2 2026):** high-throughput instrument, up to 10 billion reads per run; ~$100 genome claimed
- **PacBio SPRQ-Nx (May 2026):** HiFi human genome ~$345 list, < $300 at scale; adds 5hmC

Roche release 29 Jun 2026; Ultima (GEN, 23 Feb 2026); Element 2026 roadmap; PacBio release 26 May 2026 (all manufacturer claims)

Sequencing in Sweden: NGI

- **Illumina NovaSeq X:** short reads at scale
- **Element AVITI / AVITI24:** accurate short reads, in situ
- **PacBio Revio:** HiFi long reads
- **ONT PromethION:** ultra-long reads, direct RNA

National Genomics Infrastructure (SciLifeLab): Stockholm and Uppsala nodes. Ask them for advice on study design before you sequence: ngisweden.scilifelab.se

NGI Sweden methods and technologies pages (accessed Oct 2026)

Learn more

- Shendure J et al. (2017) DNA sequencing at 40: past, present and future. Nature 550:345–353
- Satam H et al. (2023) Next-generation sequencing technology: current trends and advancements. Biology 12:997 (open access; correction: Biology 2024, 13:286)
- Marx V (2023) Method of the year: long-read sequencing. Nature Methods 20:6–11
- Höjer P et al. (2026) Whole-genome sequencing with AVITI and NovaSeq X Plus reveals comparable performance with contextual biases. NAR Genom Bioinform 8:lqag053
- Stephens ZD et al. (2015) Big Data: Astronomical or Genomical? PLoS Biol 13:e1002195
- Goodwin S et al. (2016) Coming of age: ten years of next-generation sequencing technologies. Nat Rev Genet 17:333

Image credits

- Own work (CC BY 4.0): sample to library, pulsed-field gel, PacBio zero-mode waveguide, flow cell, Sanger workflow, gel and trace, chain termination, fragmentation, tagmentation, RNA library, Illumina SBS, two-colour chemistry, read types, PacBio HiFi, nanopore, isoforms, molecule count, read-length and N50 figures
- Cost per genome: data NHGRI via Our World in Data (CC BY 4.0)
- Growth of sequencing: Stephens et al. 2015, PLoS Biol (CC BY 4.0)
- NIPT panel: Shendure et al. 2017, Nature (© Springer Nature)